

GENERAL SPECIFICATION G2.14

Plant, Equipment and Materials (Delivery, Storage and Installation)

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G2.14 PLANT, EQUIPMENT AND MATERIALS (DELIVERY, STORAGE AND INSTALLATION)

G2.14.1 Delivery and Storage of Plant – Pre-Delivery

- (a) Prior to delivery, the Contractor shall provide the S.O. notice of shipment at least 30 days prior to shipment. This advance notification shall be submitted to the S.O. together with the following information:
 - (i) Manufacturer's instructions for handling, storage, maintenance and protection during storage and installation;
 - (ii) Assembly instructions for parts shipped and unassembled; and
 - (iii) Details of storage in Singapore if stored off Site. Details shall include location, capacity of storage facilities and method of storing the plant and equipment.

G2.14.2 Packaging and Marking

- (a) Upon the successful completion of the prescribed off-site testing and inspection and prior to dispatch from the place of manufacture, all items of equipment shall be thoroughly protected as appropriate against corrosion and incidental damage, including effects of insects, vermin, strong sunlight, rain, frost, high temperatures, humid and salty atmosphere, sea spray and the ingress of dust and sand. Packaging shall be carried out to BS 1133.
- (b) All items intended to be painted shall receive the treatment specified and all unpainted parts liable to corrosion (except for the electrical equipment) shall be thoroughly coated with a high melting point grease or tallow (free from acid) or other approved protective substance, which is easily removable during or after erection.
- (c) In addition, all electrical equipment shall be suitably protected against corrosion and incidental damage to the satisfaction of the S.O.. Temporary leads shall be fitted to electrical equipment to enable anti-condensation heaters to be energised when the equipment is in store, and such heat shall be applied by the Contractor when climatic conditions so require.
- (d) All items shall be packed to withstand land and sea transportation, and rough handling in transit. Packages shall be suitable for exposure to the atmosphere and storage in the open, including possible delays on exposed quaysides. The Contractor shall be held responsible for an item being packed so that it reaches its destination intact and undamaged. The Contractor shall provide and include for the cost of all necessary packings, cases and crates properly strengthened by battens which shall be considered as non-returnable. Packages containing dangerous or breakable goods shall be packed and marked in accordance with any statutory rules and orders applicable locally, abroad and at sea. All crates and packages shall be correctly and adequately marked as follows:
 - (i) The Board's name;
 - (ii) Title of scheme;
 - (iii) Title of contract;
 - (iv) Contractor's name;

- (v) Title of work location;
- (vi) Designation and item number identical with approved schedules; and
- (vii) Usual commercial markings.
- (e) Spare parts and special tools shall be marked to identify the associated products by name, equipment and part number. Parts shall be packaged for protection against damage from the elements during shipping, handling and storage. They shall be shipped in boxes or containers marked to indicate the content in addition to the information as stated above. Spare parts and tools shall be delivered before the associated plant or equipment is scheduled for Site testing.
- (f) The Contractor shall keep the S.O. fully informed of the status of deliveries of items, and shall submit through the formal submittal process copies of all shipping documents to the Board and to the S.O.. All shipping documents shall be accompanied by copies of test certificates as required.

G2.14.3 Delivery

- (a) The Contractor shall be responsible for the delivery of items to its site, including their handling and safekeeping. Delivery of items shall comprise all or any of the following, as may be necessary in the particular case, but not be limited to:
 - (i) Obtaining the necessary import permits for all equipment from relevant Authorities;
 - (ii) Loading and transporting items by sea, land or air freight, including insurances;
 - (iii) Taking charge of items at the port of entry or rail or air terminal in Singapore, including payment of port charges, wharfage, demurrage, intermediate storage, off-loading at terminal points, reloading and transporting to the Site; and
 - (iv) Taking delivery of items at the Site, off-loading and temporarily storing in store buildings or under cover until they are required for installation in the Works.
- (b) The Contractor shall examine items to the extent possible at the port or air or rail terminal, and more thoroughly whilst taking delivery at the Site. Should the Contractor discover that any item of Equipment has been damaged or impaired in transit, he shall forthwith direct the attention of the S.O. thereto, who will inspect and give instructions for its disposal. Items with short shelf-life such as chemical reagents shall be delivered in the shortest possible time before use with the Equipment.
- (c) Any item condemned by the S.O. shall be immediately removed and replaced by the Contractor.

G2.14.4 Storage on Site

- (a) The Contractor shall note that an area has been allocated as indicated in the Drawings as the Contractor's site office and laydown area.

- (b) If at any point of time, during the Contract Period, delivery of materials, plant or equipment create a need for additional storage space, the Contractor shall source for these storage space outside this Site on his own. All charges such as rentals, including processing / application fees by SLA and / or any other agencies for TOL land where applicable in relation to the usage of the land during the period, for completion of the Works shall be borne by the Contractor and deemed included in the rates and prices. This shall include any subsequent extensions to the period of works for off – Site storage and costs of setting up storage facilities.
- (c) The Contractor shall provide all facilities for the safe and proper storage of all Equipment including electrical and electronic components as recommended by the manufacturers, with particular consideration given to temperature, rain, humidity sunlight, wind and ground conditions.
- (d) Stored items shall be laid out by the Contractor to facilitate their retrieval for use in a programmed order.
- (e) Stacked items shall be protected from damage by spacers on load distributing supports and shall be safely arranged. No metalwork shall be stacked directly on the ground.
- (f) Small items shall be held in suitable bins, boxes or racks and shall be clearly labelled.
- (g) Items shall be handled and stored so that they are not subjected to excessive stresses and so that their protective coatings are not damaged.
- (h) The Contractor shall comply with manufacturers' package and equipment markings concerning the use and location of lifting slings, chains and hooks.

G2.14.5 Storage in Singapore

- (a) All equipment stored in Singapore prior to delivery to Site shall be stored in a properly constructed storage area approved by the Singapore Civil Defence Force, and shall be well protected from moisture. All equipment shall be supported clear of the floor by timber or pallets.
- (b) Equipment shall be stored in such a way that each crate is readily accessible and the contents of the crate can be identified without moving the crates. The storage area shall be approved by the S.O. prior to receiving equipment for storage.
- (c) All equipment shall be in Singapore at least 3 months before the programmed commencement of installation of the works. The Contractor will not be reimbursed for any of the costs of storage of the equipment if it arrives in Singapore during the 3 months period before commencement of installation.
- (d) Upon delivery to Site, but prior to actual installation, all equipment, including those provided by the nominated sub-contractors, shall be stored in a storage area provided by the Contractor, and approved by the S.O..

G2.14.6 Erection of Plant and Equipment - General

- (a) All equipment shall be erected in a neat and workmanlike manner on the foundations and at the locations shown on the approved drawings.

- (b) The Contractor shall be responsible for setting-up and erecting the Plant to the line and level required and shall ensure that all plant is securely held and remains in correct alignment before, during and after grouting or building in. This responsibility shall not be passed to any other contractor.
- (c) Where the Contractor is required to make holes and openings in existing building structures required for fixing conduits, ducting, small bore pipes and other similar small plant items including associated cast in anchor bolts, plinth and other supports, the Contractor shall submit these plans to the S.O. for approval.

G2.14.7 Equipment Foundations, Plinths and Pads

- (a) The Contractor shall be responsible for the construction of all equipment foundations, plinths and pads for all plant and equipment such as pumps, valves, tanks, panels etc supplied and installed under this Contract.
- (b) Starter bars for foundations, plinths and pads shall be constructed based on the layouts and dimensions given by the Contractor, in accordance to the equipment, valves, pumps etc to be installed. The Contractor shall carry out the installation and carry out Site verification to confirm the accuracy and location of the starter bars to confirm its adequacy.
- (c) Foundations shall comply with the details shown on the approved drawings. Where required, the Contractor shall submit details of pockets, sleeves, holding down bolts, concrete bases, etc required for installation of his plant and equipment. Where floors and plinths are required to be cut away and prepared to accept foundation bolts or mounting frames and plates, S.O.'s approval shall be sought and obtained prior to the commencement of any works.
- (d) Prior to the casting of structures and fixing of foundation bolts, plinths and fixings, the Contractor shall satisfy himself on-Site that the locations and dimensions required for fixing of foundations, including cast in anchor bolts, plinths and other supports are carried out per the design and equipment selection. The Contractor shall carry out a joint inspection with the S.O. prior to the casting.
- (e) Should the Contractor require changes or add to any floors or plinths after they have been constructed, the Contractor shall bear all costs associated with procuring new foundation bolts and fixings, demolition, rectifying, modifying and re – construction of the floors and plinths including fixing of the foundation bolts and fixings.
- (f) Where the installed starter bars are off – alignment after verification and casting, the Contractor shall modify by cutting, and installing additional dowels where necessary at his own cost to facilitate the construction of the required plinths or pads.

G2.14.8 Forming of Openings in Structures

- (a) The Contractor shall provide exact locations and dimensions of all openings needed to be formed in walls and floors through which services supplied and installed under this Contract are to penetrate.
- (b) The Contractor shall be on-Site to confirm and coordinate the exact locations of openings to be provided to meet his requirements.

- (c) Where the formed openings are found to be off –alignment or incorrect in dimensions or locations, or if any other openings or holes or additional loads applied to the structures are found to be required, or if the Contractor wishes to make any changes, subsequent to approval of the drawings prepared by the Contractor to accommodate the Plant and the issue of final construction drawings, all such modifications to existing and / or additional holes or openings shall be cut by the Contractor at his own expense. Locations and sizes and method of cutting shall be submitted by the Contractor to the S.O. for approval.
- (d) The Contractor shall note that in this case approval will be dependent on an assessment by the S.O. on whether the structural integrity of the building or structure will be compromised, or whether they will be detrimental to the appearance of the finished waterworks. If the S.O. should refuse permission to cut such holes or openings, or to apply additional loads to the structures or buildings for these reasons, the Contractor shall make fresh proposals, and shall have no claim for delay in this event.

G2.14.9 Sealing of Openings after Installation of Services

- (a) After installation of services through openings in the floors and walls, the Contractor shall seal all openings in the structures.
- (b) All annular spaces between pipes, openings in walls and floors shall be roughened and grouted in after installation of services. The Contractor shall be responsible for the water – tightness of all structures at the grouted in portion of the wall or floor.
- (c) For cabling and ductworks, the annular space between the openings in floors or walls and services shall be sealed using the approved type of sealing compound to meet the required zone (fire) separation requirements between the rooms.

G2.14.10 Foundation Bolts and Fixings

- (a) All plant and equipment supplied and delivered under this Contract shall be installed firmly in position by the Contractor with holding or connection bolts on concrete bases.
- (b) All anchor bolts, holding down bolts, fixing bolts, washers, nuts, straps, supports, brackets, spacers and fixtures, which are necessary to be cast into the structure or grouted for the satisfactory installation and erection and installation of the Plant shall be supplied by this Contract.
- (c) The Contractor shall provide all necessary design details and specifications to the S.O. for all anchor bolts (cast in situ or grouted), holding down bolts, fixing bolts, washers, nuts, straps, supports, brackets, spacers, shims and fixtures, which are necessary for the satisfactory installation and erection and installation of the Plant.

G2.14.11 Fixing to Structures

- (a) The Contractor shall be responsible for supplying, installing and building in cast in place items such as conduits, pipes, pipe sleeves and other similar items into the structures, including forming openings in the structures where required for all plant and equipment supplied and installed under this Contract.

- (b) All cast in items in water retaining structures shall be built in during concreting works. Cast in items shall be provided with puddle flanges and installed as detailed on the Drawings. The Contractor shall modify the formwork, and reinforcement to accommodate the embedded items and openings.
- (c) No holes in concrete, brick, timber or structural steelwork for fixing devices shall be drilled without the prior approval of the S.O.. Holes shall be only be made with a core drilling, saw drill, rotary drill of the non-percussion type. The use of jack hammers or other percussion type of equipment to form holes in the structures is not allowed. A single shot cartridge tool may be used only at the discretion of the S.O.. All drill in / chemically fastened bolts, including grouting of base plates shall be supplied and installed by the Contractor.
- (d) Any damage caused by the Contractor during erection to any surface shall be made good by the Contractor at his own expense to the satisfaction of the S.O..
- (e) The Contractor shall be responsible for the provision of any scaffolding required and the cost of supplying this shall be deemed to be included in the rates and prices.

G2.14.12 Installing Embedded Items

- (a) Unless specified otherwise, the Contractor shall supply, handle and install in final positions all embedded items such as pipes, conduits, pipe sleeves, and other similar items. All items shall be ready and built into the structure.
- (b) The Contractor shall incorporate all embedded items into his construction and fabrication coordination drawings.
- (c) The Contractor shall provide the necessary arrangements and work areas, including all scaffolding arrangements for installation of these embedded items. The Contractor shall provide all required lifting, support and any other equipment necessary to move the embedded items into its final place.
- (d) The Contractor shall be responsible to rectify any embedded items which have not been properly built in. Prior to proceeding with any rectification work, the Contractor shall submit his method of rectification to the S.O. for approval. All rectification work or re – building in of the embedded items, including grouting work shall be carried out by the Contractor and deemed included in the Contract sum.

G2.14.13 Plant Protection on Site

- (a) Factory finished and Site painted Plant shall be adequately protected during and after installation against damage to finished surfaces and fitted components and against ingress of dust, moisture and the effects of high humidity. It may be necessary for structural and building finishing operations to be carried out near installed plant and the Contractor shall take this into consideration in complying with the requirements of this Clause. The cost of all protection to installed plant as required by the S.O. shall be deemed to be included in the rates and prices.

G2.14.14 Cleaning and Preparing Surfaces at Site

- (a) Immediately on their arrival at the Site, the Contractor shall arrange for all items of Plant to be inspected by the Contractor and the S.O. for damage to protective coatings applied at the place of manufacture. Damaged portions shall be cleaned down, all rust removed by grit blasting or other surface preparation process, if instructed by the S.O., and the coating made good with a similar paint within four hours of surface preparation.
- (b) After erection and building in, appropriate items of Plant shall be thoroughly cleaned to remove all debris, dust and grease and filled where necessary.

G2.14.15 Protective Coatings Applied at Site

- (a) The Contractor shall apply the protective coatings for the Plant at Site strictly in accordance with the detailed protective treatment system approved by the S.O..
- (b) The first in-situ coat shall in all cases be applied immediately after the items of Plant have been erected and built in.
- (c) The painting of all Plant shall be completed as a condition precedent to the issue of the Completion Certificate.
- (d) Before the end of the Defects Liability Period, all Plant shall have the paint work touched up, and the cost of this shall be deemed to be included in the Contract Price. Where touch up is carried out, the end product shall display uniformity of colour with the original paint coating. If uniformity of colour is not achieved after touching up, to the S.O.'s satisfaction, then the entire Plant item shall be re-painted such that uniformity of colour is achieved and the cost of this work shall be deemed to be included in the rates and prices.

G2.14.16 Connection of Electrical Power

- (a) The Contractor shall check all items of electrical plant for correct phasing and insulation resistance. Motors shall be dried out in accordance with the manufacturer's instructions and checks on the insulation resistance shall be made at regular intervals. Drying out shall continue until the insulation resistance reaches a steady value as recommended by the manufacturer. After all drying out and checking of insulation resistance is complete, the Contractor shall check that all electrical connections to the plant are correct, any errors shall be brought immediately to the notice of the S.O..
- (b) Only after the individual tests for electrical plant, as specified, have been carried out to the satisfaction of the S.O., shall electric power be applied to any plant. First-time energising of main plant shall not be carried out unless witnessed by the S.O. and the Board's representative.

G2.14.17 Construction Equipment and Methods

- (a) The Contractor's Construction Equipment shall be appropriate for the expected duties and shall be deployed in sufficient numbers to achieve a satisfactory quality of work and a rate of progress which ensure the completion of the Works within the duration stated for each milestone as shown in the Contract.

- (b) If at any time such Construction Equipment appears to the S.O. to be inefficient, inappropriate or insufficient for securing the quality of work required or for producing the required rate of progress, the S.O. will notify the Contractor in writing and the Contractor shall take such steps as are necessary to achieve the quality of workmanship and/or expedite the progress. Failure of the S.O. to give such notification shall in no way relieve the Contractor of his obligations to secure the quality of the work and rate of progress required.
- (c) The Contractor shall use all reasonable means to prevent noise and annoyance to the inhabitants of the area in which the Works is situated, and all machinery must be of such design and so arranged as to be reasonably free from noise in operation. Where air compressors or pneumatic tools are being used in the Works, the Contractor shall arrange, when required by the S.O. for such compressors and tools to be fitted with silencers of approved design or he shall adopt other approved means of reducing the noise. The Contractor shall maintain and operate all fuel burning plant and equipment such that the requirements of local law as and regulations are complied with. No claim by the Contractor for complying with the provisions of this Clause shall be entertained by the Board. The Contractor shall comply with the noise requirements specified in the HSE specifications.
- (d) The Contractor is deemed to be fully aware of the local authority requirements and regulations on the usage and deployment of plants and machinery. In this respect, the requirements in particular the requirements of National Environment Agency (NEA) on pollution control, the requirements of Building Control Authority (BCA) on vibration control and impact on existing buildings and structures, the requirements of Energy Market Authority (EMA) on usage of power generating equipment, the requirements of Ministry of Manpower (MOM) on safety, use of pressurise equipment, the requirements of service providers on protection of existing services etc amongst others shall be given due consideration when pricing for the project.
- (e) If in the S.O.'s opinion the Contractor's failure to comply with the above requirements make it desirable that the use of the Construction Equipment should be suspended either temporarily or permanently, the Contractor shall change the method of performing the work affected at his own cost. He shall be deemed to have no cause for claim against the Board on account for having to carry on the work by another method nor shall be deemed to have cause for claim if any order issued by the S.O. results in the Construction Equipment having to stand idle for a period of any duration whatsoever or having to be removed.
- (f) The Contractor shall implement a preventive maintenance programme for the Construction Equipment. Further requirements are specified in the HSE specifications.

END OF SECTION